

## Rosen — §2.2: Set Operations

### Selected Exercises

Alston

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#### Exercise 15

Prove the second De Morgan law in Table 1 by showing that if  $A$  and  $B$  are sets, then  $\overline{A \cup B} = \overline{A} \cap \overline{B}$

(a) by showing each side is a subset of the other side.

(b) using a membership table.

#### Exercise 17

Show that if  $A$ ,  $B$ , and  $C$  are sets, then  $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$

(a) by showing each side is a subset of the other side.

(b) using a membership table.

#### Exercise 19

Show that if  $A$  and  $B$  are sets, then

(a)  $A - B = A \cap \overline{B}$ .

(b)  $(A \cap B) \cup (A \cap \overline{B}) = A$ .

#### Exercise 31

Let  $A$  and  $B$  be subsets of a universal set  $U$ . Show that  $A \subseteq B$  if and only if  $\overline{B} \subseteq \overline{A}$ .

#### Exercise 37

Show that if  $A$  is a subset of a universal set  $U$ , then

(a)  $A \oplus A = \emptyset$ .

(b)  $A \oplus \emptyset = A$ .

(c)  $A \oplus U = \overline{A}$ .

(d)  $A \oplus \overline{A} = U$ .

#### Exercise 41

Suppose that  $A$ ,  $B$ , and  $C$  are sets such that  $A \oplus C = B \oplus C$ . Must it be the case that  $A = B$ ?

#### Exercise 51

Find  $\bigcup_{i=1}^{\infty} A_i$  and  $\bigcap_{i=1}^{\infty} A_i$  if for every positive integer  $i$ ,

- (a)  $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$ .
- (b)  $A_i = \{-i, i\}$ .
- (c)  $A_i = [-i, i]$ , that is, the set of real numbers  $x$  with  $-i \leq x \leq i$ .
- (d)  $A_i = [i, \infty)$ , that is, the set of real numbers  $x$  with  $x \geq i$ .